

Ejer identidad auxiliar disco unidad

Ejercicio 1. Sean $\alpha, \beta \in \mathbb{D}$. Demuestra la siguiente identidad:

$$1 - \left| \frac{\alpha - \beta}{1 - \bar{\beta}\alpha} \right|^2 = \frac{(1 - |\alpha|^2)(1 - |\beta|^2)}{|1 - \bar{\beta}\alpha|^2}.$$

Solución: Partiendo del lado izquierdo, tenemos que

$$1 - \left| \frac{\alpha - \beta}{1 - \bar{\beta}\alpha} \right|^2 = \frac{|1 - \bar{\beta}\alpha|^2 - |\alpha - \beta|^2}{|1 - \bar{\beta}\alpha|^2}.$$

En el numerador, desarrollando ambos términos, obtenemos

$$\begin{aligned} |1 - \bar{\beta}\alpha|^2 - |\alpha - \beta|^2 &= (1 - \bar{\beta}\alpha)(1 - \beta\bar{\alpha}) - (\alpha - \beta)(\bar{\alpha} - \bar{\beta}) \\ &= 1 - \bar{\alpha}\beta - \alpha\bar{\beta} + |\alpha|^2|\beta|^2 - (|\alpha|^2 - \alpha\bar{\beta} - \bar{\alpha}\beta + |\beta|^2) \\ &= 1 - |\alpha|^2 - |\beta|^2 + |\alpha|^2|\beta|^2 = (1 - |\alpha|^2)(1 - |\beta|^2). \end{aligned}$$

Por tanto, se sigue la identidad deseada. ■

Referenciado en

- `ejer-desigualdad-schwarz-pick`

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